



Real State Investment Advisor Dashboard Insights

By Vishal Kr. Mohali

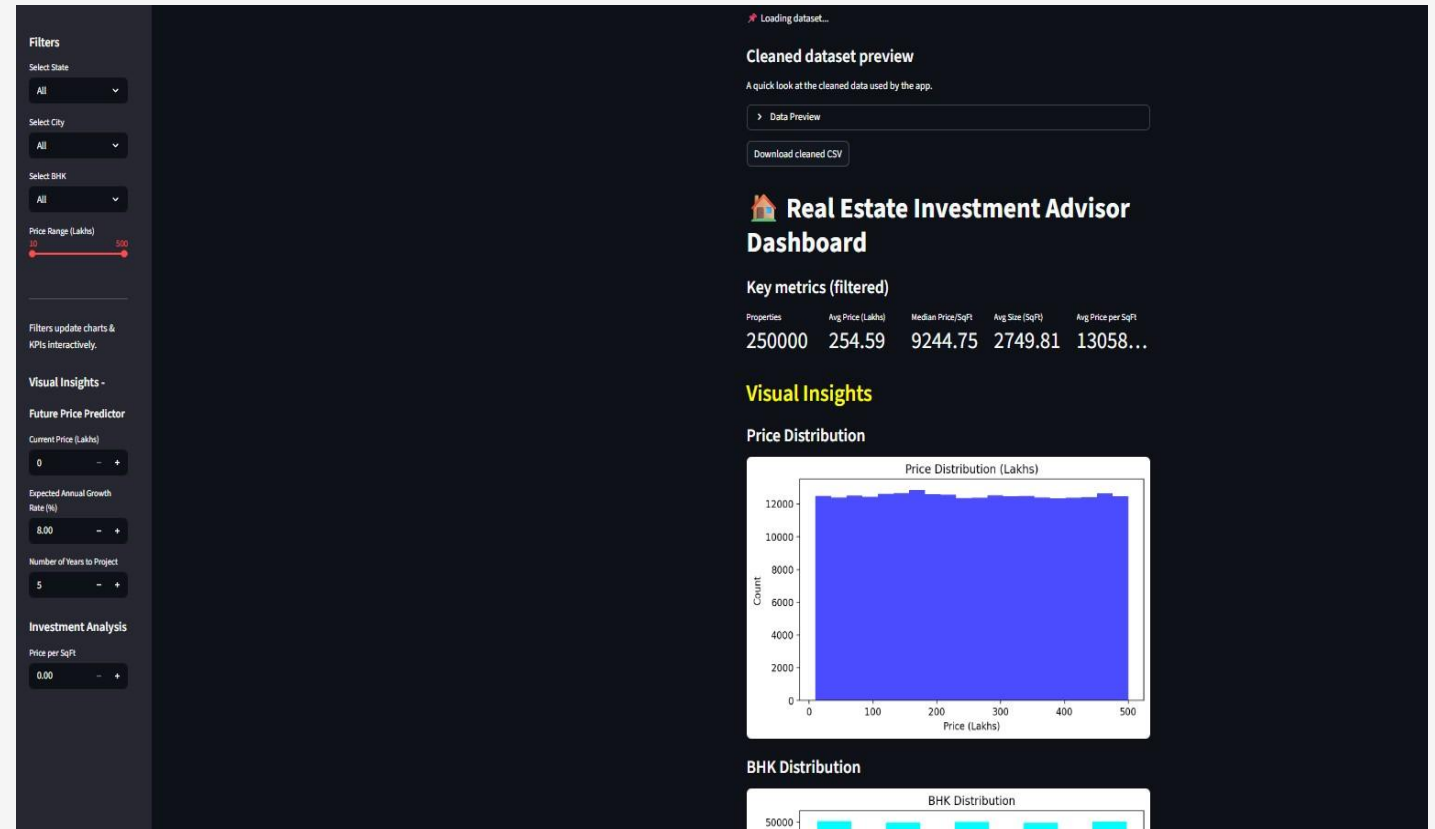
INTRODUCTION

Hello, In this presentation, we will go through the **Real Estate Investment Advisor: Predicting Property Profitability & Future Value Dashboard**.

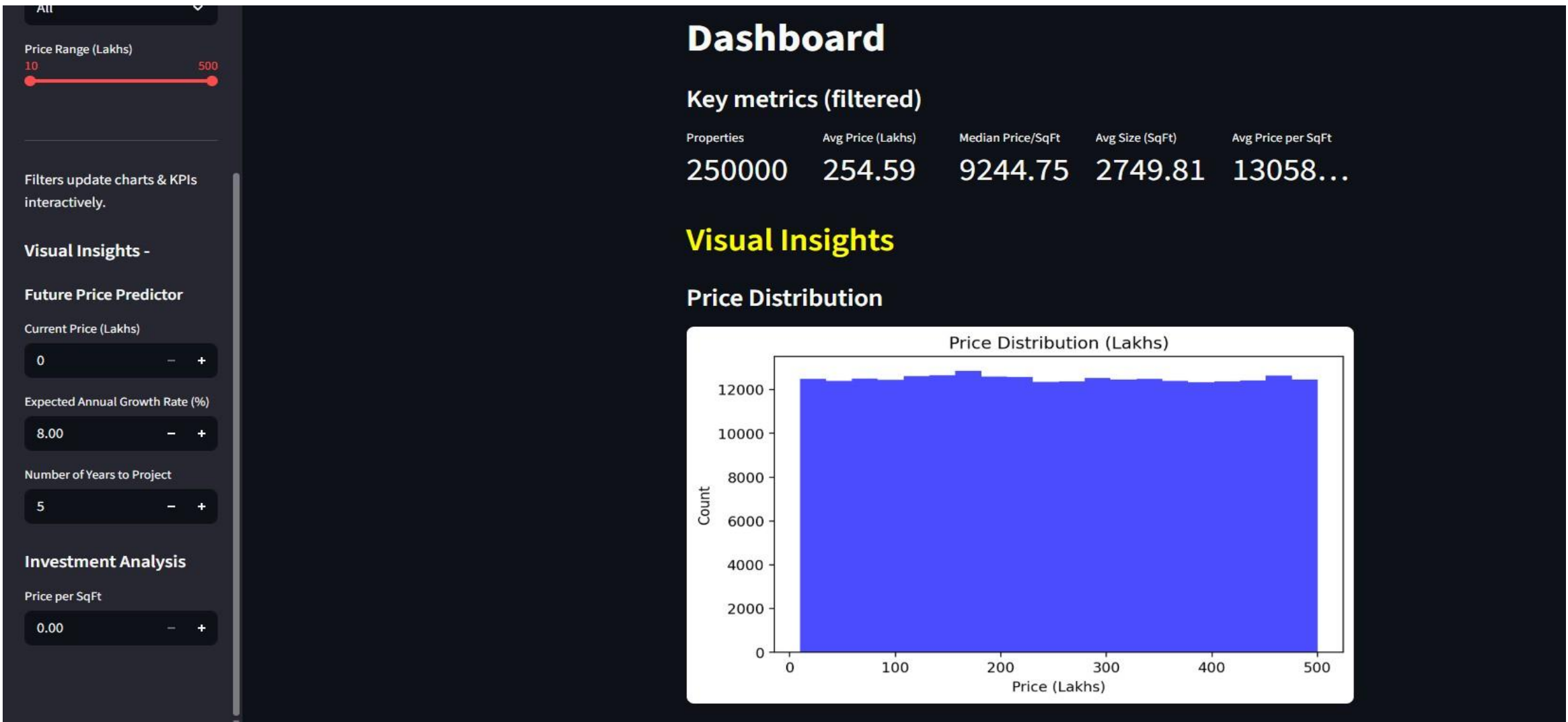
- I appreciate the opportunity given to me to dive into this Real Sate Investment Advisor data to gain insightful information about the business performance.
- And How to think from the perspective of business leaders to analyse data more effectively.

Since ML was not used, all predictions are rule-based, transparent, and easy to interpret. The project includes three major parts:

Data Cleaning, EDA and Streamlit Dashboard for Insights & Investment Evaluation."



- This Streamlit dashboard helps stakeholders evaluate regional performance, monitor growth trends, that provides investment recommendations and price forecasts, and derive business insights essential for strategic decision-making.
- This Provide rule-based investment guidance and future price estimates using the provided dataset.



PROCESS

All the necessary steps were taken in Python Code to ensure that the analysis is accurate and correct.

Preprocessing Steps:

- Removed duplicates (if any)
- Filled numeric missing values with median
- Filled categorical missing values with mode
- Numeric type conversion
- Price_per_SqFt and Age_of_Property creation
- Recomputed ``Price_per_SqFt`` = $\text{Price_in_Lakhs} * 100000 / \text{Size_in_SqFt}$
- Created ``Future_Price_5yrs`` using an 8% annual growth rate
- Created ``Good_Investment`` label comparing property PPSQFT to city median
- Outlier detection using IQR
- Trend analysis and heatmaps

EDA Findings:

- Top cities by average price: (see notebook)
- Price per sqft varies strongly by city and property type.
- Amenities, parking and nearby schools show correlation with price_per_sqft
- Price trends by city
- Correlation between area and investment return
- Relationship between infrastructure score and resale value

Dashboard Layout

In the Streamlit Dashboard, the layout is designed to provide a comprehensive overview of the **Real Estate Investment Advisor: Predicting Property Profitability & Future Value** data at a glance.

The KPI cards prominently display key metrics, including the total Properties, Average Price, median price/sqft, Average size in sqft, Average Price/Sqft, and On the right side, I included **Filter with Dropdown slicers and Custom Mode Filtering for Visual Insights** which enable to predict future value.

At the bottom, I added **Custom Mode Evaluation Metrics (Calculator)** like: States, City and by BHK, Yearly growth rate, Current Price, Size(Sqft), Property Availabe or Not, Has Parking or Not.

In Visual's We explore price trends using histograms, observe size–price relationships using scatter plots, and identify outliers using statistical thresholds. Heatmaps and bar charts reveal city-level and state-level variations.

These metrics deliver essential insights into the market's health and ensures that users can explore the dataset with complete flexibility.

Dashboard

Key metrics (filtered)

Properties	Avg Price (Lakhs)	Median Price/SqFt	Avg Size (SqFt)	Avg Price per SqFt
250000	254.59	9244.75	2749.81	13058...

Filters

Select State

All

Select City

All

Select BHK

All

Price Range (Lakhs)

10

500

Filters update charts & KPIs interactively.

Visual Insights -

Future Price Predictor

Current Price (Lakhs)

0

Expected Annual Growth Rate (%)

8.00

Number of Years to Project

5

Investment Analysis

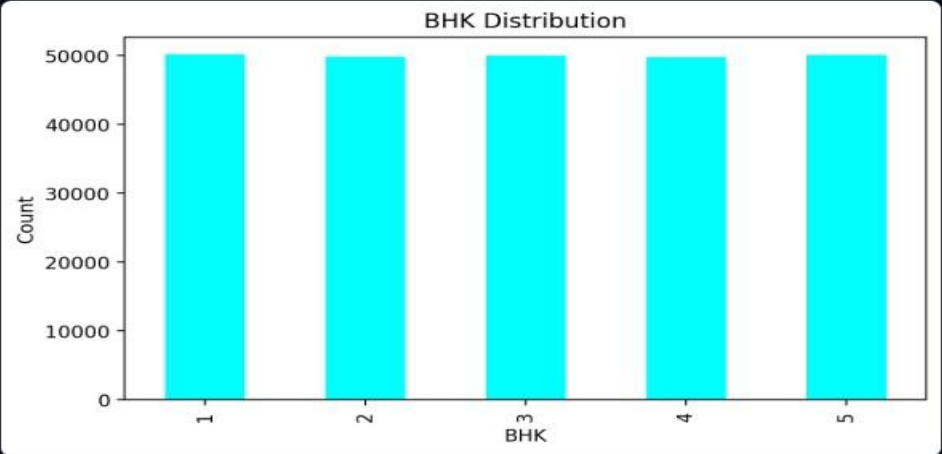
Price per SqFt

0.00

Visual Chart Analysis

The visuals in this dashboard provide a deep understanding of location-wise heatmaps, trend charts.

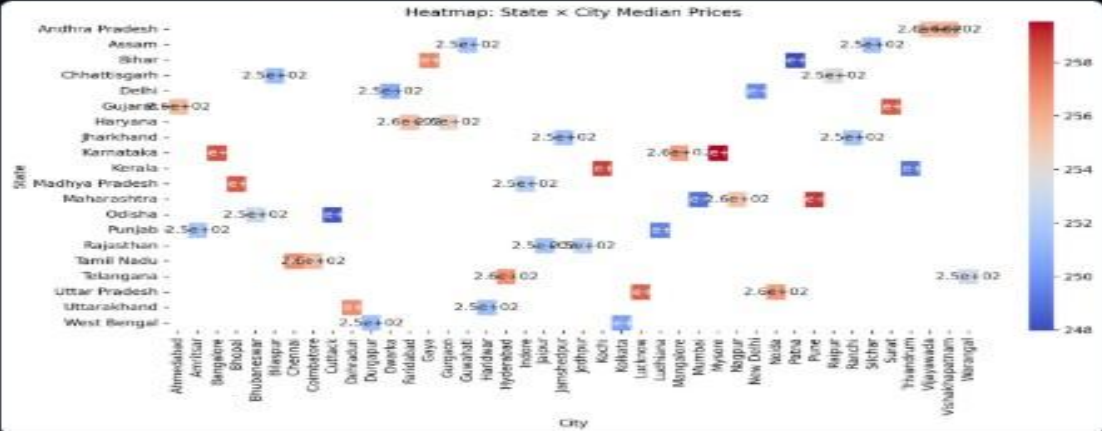
BHK Distribution



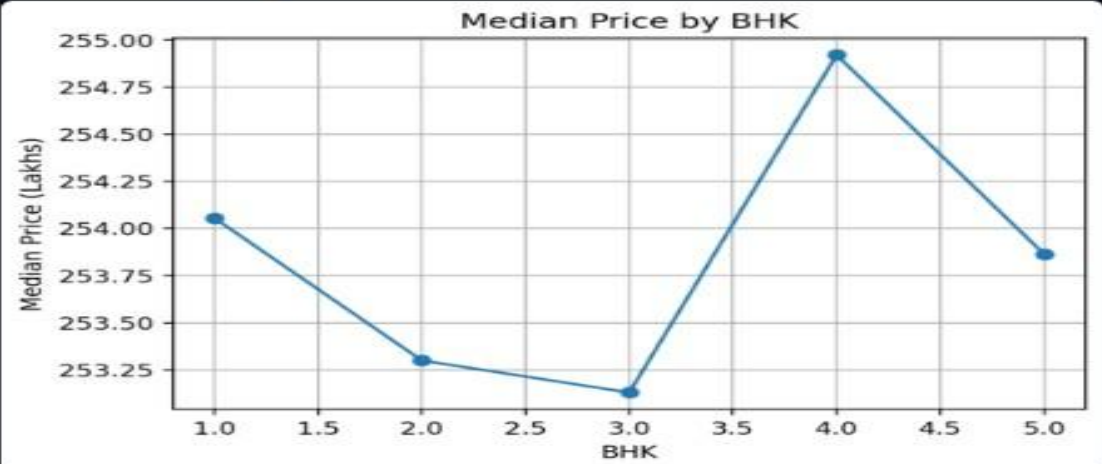
Average Price by City



Heatmap: Median Price by State & City



Median Price by BHK



Business Questions Answered by the Dashboard

The dashboard is designed to help investors, analysts, and homebuyers make informed decisions about property investments. It addresses the following core business questions:

Market Overview:

- What is the distribution of property prices across different states and cities?
- How do property sizes and BHK configurations vary in the dataset?
- Which states and cities have the highest concentration of properties?

Investment Evaluation:

- Is a given property a **good investment** compared to local medians?
- What is the **estimated future price** of a property after 5 years, given growth assumptions?
- How does the property's **price per square foot** compare to city/state medians?

Trend & Comparative Analysis:

- How do median prices vary by BHK configuration?
- Which cities show higher average prices, and how do they compare across states?
- What are the **outliers** in property prices (extremely high-value properties)?

Feature Importance & Confidence:

- Which property features (size, BHK, parking, RERA, etc.) most strongly influence price per square foot?
- How confident is the classification decision (rule-based confidence score)?

Business Use Cases

- ✓ Empower real estate investors with intelligent tools to assess long-term returns.
- ✓ Support buyers in choosing high-return properties in developing areas.
- ✓ Help real estate companies automate investment analysis for listings.
- ✓ Improve customer trust in real estate platforms with data-backed predictions.

Key Insights & Findings

From the dashboard visuals and evaluation metrics:

Price Distribution:

- Property prices are concentrated in the lower to mid-range, with a long tail of high-value outliers.
- Outlier detection highlights luxury properties that skew averages upward.

BHK & Size Trends:

- Median prices increase with higher BHK counts, but the rate of increase is not linear.
- Larger properties tend to have higher absolute prices, but price per square foot often stabilizes beyond a certain size.

City & State Comparisons:

- Cities like Mumbai, Bengaluru, and Delhi show consistently higher average prices.
- Tier-2 cities (e.g., Lucknow, Jaipur, Coimbatore) present more affordable options with moderate growth potential.
- State-wise counts reveal concentration in Maharashtra, Karnataka, and Delhi NCR.

Investment Analysis:

- Properties priced below the **city median** or with **lower price per square foot** are flagged as "Good Investment."
- Multi-factor scoring (BHK $\geq$ 3, RERA compliance, ready-to-move, parking availability) strengthens investment classification.
- Example evaluation showed a property with **Price per SqFt = ₹25,387** compared to a **city median of ₹9,244**, resulting in a "Not Good" classification.

Future Price Prediction:

- Using compound growth (e.g., 8% annually), a property priced at ₹253.87 Lakhs today is projected to reach ~₹373 Lakhs in 5 years.
- City-specific growth rates adjust forecasts upward for metros and slightly lower for smaller cities.

Feature Importance:

- Strongest correlations with price per square foot were found in:
- **Size_in_SqFt** (positive correlation)
- **Price_in_Lakhs** (expected strong correlation)

Other features like BHK, nearby schools, and parking showed weaker but notable influence.

Suggestions & Recommendations

Based on the dashboard's insights, here are actionable recommendations:

For Investors:

- Focus on properties priced **below local medians** with favorable multi-factor attributes (≥ 3 BHK, RERA, ready-to-move, parking).
- Use the **future price predictor** to estimate long-term appreciation and compare across cities before committing capital.
- Avoid overpaying for properties with price per square foot significantly above city averages.

For Analysts:

- Enhance the dashboard with **time-series data** (year-on-year price trends) to capture market cycles.
- Incorporate **location clustering** (metro vs tier-2 vs tier-3 cities) for clearer segmentation.
- Expand feature importance analysis to include **amenities, infrastructure projects, and proximity to business hubs**.

For Homebuyers:

- Use filters interactively to shortlist properties by budget, size, and city.
- Pay attention to **classification confidence scores**—low confidence suggests further due diligence is needed.
- Consider long-term affordability by checking projected future prices against expected income growth.

For Dashboard Improvement:

- Add **interactive maps** for geographic visualization of property clusters.
- Provide **scenario simulators** (e.g., growth rates at 6%, 8%, 10%) to test sensitivity of future price predictions.
- Include **downloadable reports** summarizing filtered insights for offline review.

Summary

The dashboard successfully answers critical investment questions, highlights market trends, and provides rule-based evaluations without complex ML models. It empowers both novice and experienced users to make data-driven property decisions with clarity and confidence.

Thank you for watching my presentation...